

DATA 266: Generative AI, Fall 2026

Lab 1 Report

Character-Level Language Model Pretraining, Sentiment Classification,
and CycleGAN Image Style Transfer

Team 09 (PairProgramming_Team_09)

Shibin Thomas

Denisha Ketan Tank

Repository: <https://github.com/Shibin-1020/Gen-AI---LAB-1>

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1 Team Ownership Statement

Shibin Thomas built the character-level GPT for Task 1, the mean-pooling baseline, TextCNN and BiGRU-with-attention classifiers for Task 2, and the CycleGAN for Task 3, including the Kaggle submission. Shibin also set up the shared TinyStories download, the shared metric modules for Tasks 1 and 2, the repository layout, the run manifests and the smoke tests.

Denisha Ketan Tank independently built and trained her own Task 1 character-level language model, three Task 2 sentiment classifiers, and her own Task 3 CycleGAN in the member_denisha folders. Denisha added the corresponding configurations, notebooks, checkpoints, metrics, plots, samples, manifests, raw logs, failure analyses, and submission artifacts. Each member trained their own models from scratch for every task.

To keep the results comparable, the team shared the raw TinyStories file and generation prompts for Task 1, the official Yelp Polarity test split and common metric module for Task 2, and the instructor's evaluation script for Task 3. The comparison tables and analysis sections of this report were written jointly.

1.1 Repository Organisation and Reproducibility

Every member folder follows the same layout: source code and the task notebook in src/, configuration files in configs/, model weights in checkpoints/<run_id>/, plots and samples in outputs/<run_id>/, and the files metrics_report.csv, results.md and failure_analysis.md. Unedited training logs are kept in reproducibility/raw_logs/ and a manifest for each run (configuration, git commit, library versions, hardware and checkpoint hashes) in reproducibility/manifests/. Each figure and table below names the file it was taken from, so every number can be traced back to its log or checkpoint.

Denisha Ketan Tank - Contribution Overview

This contribution page supplements the original Team 09 ownership statement. Shibin Thomas's original work is retained in full; the following records Denisha's independent work under the same Lab 1 requirements.

Area	Denisha contribution
Task 1	Designed and trained a character-level decoder-only GPT from scratch on TinyStories. Implemented training, checkpointing, smoke testing, metrics, generation samples, loss curves and failure analysis.
Task 2	Trained a mean-pooling baseline, TextCNN and experimental GRU sentiment classifier from scratch on Yelp Polarity. Produced full metrics, confidence intervals, slice analysis, McNemar comparison and the required 20-case error review.
Task 3	Trained an independent CycleGAN with two generators and two discriminators. Generated translations, reconstructions, official evaluation outputs and the Kaggle submission.
Reproducibility	Maintained independent member_denisha folders, configurations, checkpoints, raw logs, manifests, results and failure-analysis files.
Shared team evidence	Both members use the shared repository: https://github.com/Shibin-1020/Gen-AI---LAB-1
Audit status	The required two-rater audit uses 30 fixed images. The image bundle and templates are present; final ratings and agreement must be recorded after two independent human reviews.

Denisha evidence is stored under task1_llm/member_denisha, task2_sentiment/member_denisha and task3_gan/member_denisha. The personal repository is not required in the report; the shared repository is the team submission location.

2 Task 1: GPT-Style Character-Level Language Model

The model is a decoder-only Transformer trained on characters from TinyStories [2]. All components (token and positional embeddings, layer normalisation, multi-head causal self-attention, the feed-forward blocks and residual connections) were written by hand following Vaswani et al. [1], with the pre-LayerNorm arrangement of Xiong et al. [5] and the GPT-2 initialisation scheme [4]; no built-in Transformer or attention module of PyTorch is used. Each member drew an independent split from the cleaned story pool and trained a separate model from scratch.

2.1 Model Comparison

Table 1. Task 1 comparison of architecture, hyperparameters and metrics. Values are taken from each member's committed metrics file and checkpoint.

Metric / setting	Shibin Thomas (gpt_char_v1)	Denisha Ketan Tank (member_denisha)
Architecture	6 layers, d_model 320, 8 heads (head dim 40), FFN 4x with GELU, pre-LayerNorm, learned positions, tied input/output embeddings, context 256	4 layers, d_model 256, 4 heads (head dim 64), FFN 1024 with GELU, learned token/position embeddings, decoder-only causal attention, context 128
Hyperparameters	AdamW (0.9, 0.95), weight decay 0.1, peak LR 0.0006 with 2% warm-up and cosine decay to 0.00006, batch 64 x 256, gradient clip 1.0, dropout 0.1, 10 epochs, bf16	AdamW (0.9, 0.999), weight decay 0.01, peak LR 0.0005 with 500-step warm-up and cosine decay to 0.00005, batch 48 x 128, gradient clip 1.0, dropout 0.1, 10 epochs, float16 AMP
Training cross-entropy (nats/char)	0.5594	0.7128
Validation cross-entropy (nats/char)	0.5711	0.6822
Validation perplexity	1.7701	1.9783
Validation bits per character	0.8239	0.9842
Training perplexity / BPC	1.7497 / 0.8071	2.0396 / 1.0288
Generalisation gap (val - train CE)	0.0116	-0.0306
Next-character accuracy (validation)	0.8173	0.7822
Distinct-1 / 2 / 3	0.2277 / 0.7048 / 0.9264 (sampled, T=0.8)	0.1080 / 0.4582 / 0.7360
Repeated 4-gram rate	0.0066 sampled / 0.1373 greedy	0.1526
Gradient norm	mean 0.304 / p99 1.145 / max 14.71	final 0.2318 / p99 not recorded / max 0.2872
Clipped steps / loss spikes / NaN steps	1.34% / 0 / 0	not recorded / not recorded / 0
Parameters	7,509,120 total / 7,398,400 non-embedding	3,250,688 total
Training throughput	569,917 characters/s	224,657.9 tokens/s
Generation speed	255.0 characters/s	268.0 characters/s
Peak GPU memory	3,347 MB	408.3 MB
Training time / epochs / best epoch	28.9 min / 10 / 10	66.6 min / 10 / 10
Hardware	NVIDIA GeForce RTX 5090	NVIDIA Tesla T4
Run and checkpoint	gpt_char_v1_20261001-153007/best_model.pt (SHA-256 97647773d06f)	gpt_from_scratch.pt (SHA-256 3858840ebe56b6d8a85cd809cff640b9ab94c0923a0cead764fccb87ee892fb0)

Sources: task1_llm/shibin_thomas/metrics_report.csv and task1_llm/member_denisha/outputs/metrics.csv. Denisha did not record clipped-step count, loss-spike count, or gradient p99 in her metrics file; those entries are explicitly marked as not recorded rather than estimated.

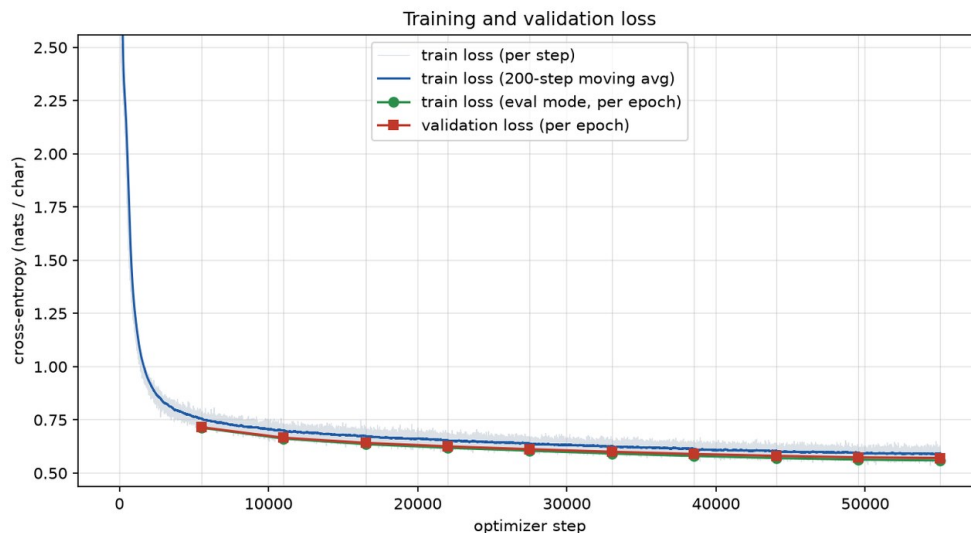


Figure 1. Task 1 training and validation loss (Shibin Thomas)
(task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007/loss_curves.png)

Table 2. Validation cross-entropy (nats/char) and next-character accuracy (%) after each epoch.

Epoch	1	2	3	4	5	6	7	8	9	10
Val. CE	0.715	0.666	0.642	0.625	0.612	0.600	0.590	0.581	0.574	0.571
Val. acc.	77.4	78.8	79.6	80.1	80.5	80.8	81.2	81.4	81.6	81.7

Source: reproducibility/raw_logs/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007/epochs.csv

The training loss starts at 4.52, which is the loss of a uniform guess over 90 characters, and drops below 1.0 within the first 1,400 steps. After that the validation loss keeps falling every epoch, from 0.715 to 0.571, so the last epoch is also the best one. The gap between validation and training loss stays very small (0.012 nats at the end), which is expected with roughly 120 training characters per parameter. Training was stable: there were no NaN steps or loss spikes, and all of the 738 clipped steps happened during the learning-rate warm-up.

2.3 Discussion

Strengths. A small model of 7.5 million parameters, trained for 29 minutes on one GPU, already writes fluent TinyStories-style text with named characters, dialogue and simple cause and effect. It reaches a validation BPC of 0.824 and predicts the next character correctly 81.7% of the time. With temperature sampling the output is varied (Distinct-3 of 0.93 and almost no repeated 4-grams).

Weaknesses. Greedy decoding falls into loops, with a repeated 4-gram rate of 0.137. Because words are produced one character at a time, rare words are often misspelled, and sometimes spelled differently each time they appear. Stories also lose their thread once they grow beyond the 256-character context.

Limitations. Each model was trained once with a single seed. The validation loss was still decreasing at epoch 10, so the model is not fully converged. Distinct-n and the repetition rate measure diversity rather than story quality, and generation speed was measured without a key-value cache.

Next steps. We would train for more epochs or use a slightly larger model, extend the context to 512 characters, and compare top-k or nucleus sampling with a repetition penalty on the shared prompts. A key-value cache would speed up generation, and a sub-word tokenizer trained from scratch would remove most of the spelling errors.

2.4 Failure Analysis (Shibin Thomas)

We generated text for the 10 shared prompts, with one greedy and three sampled continuations ($T = 0.8$) of 500 characters each. The three clearest failures are shown below; the full set of 40 samples is in `outputs/<run_id>/samples.md`.

Case 1: Repetition (greedy decoding)

In a small house near the forest, there was a big tree. The tree was very high and the tree was very high. The tree was very high up in the tree. The tree was very high and the tree was very high.

One day, the tree saw a big tree. The tree was very high and the tree was very high. The tree was very high and the tree was very high. ...

The phrase “the tree was very” appears 16 times and the repeated 4-gram rate of this sample is 0.61, against 0.04 on average. Greedy decoding always takes the most likely character, so once a high-probability phrase has been written, starting it again is again the most likely continuation. Sampling, or a repetition penalty, would break the loop; the repeated 4-gram rate on the same prompts is the measure to check.

Case 2: Broken and inconsistent spelling ($T = 0.8$)

Tim was scared. He ran and ran until he came across a big lute. He saw a small tree which was very pretty. Tim wanted to take the lutter from him. He said yes, but he had to be careful.

The luter was big and boring. Tim saw that the lutte was scary and hurt. ...

The rare word “lute” is spelled four different ways (lute, lutter, luter, lutte). The model has no word list and rebuilds a rare word from frequent letter patterns each time it comes back. A lower temperature, longer training or sub-word tokens should reduce this.

Case 3: Loss of coherence ($T = 0.8$)

After school, Sam wanted to go to the zoo and see all the animals.

They all walked closer and held on tight. Sam thought it was tremometer than the diamond. ... Together, Jack and Sam started to fill their bucket with food. He was so proud of their work! ... Tom and Sam continued to throw their bucket until it was time to go hom

Each sentence is fluent, but the setting jumps from the zoo to a diamond and a bucket, two new friends (Jack and Tom) appear without introduction, and the pronouns stop matching. The continuation is 500 characters long while the model only sees the last 256, so the beginning of the story has literally dropped out of its context. A longer context or a deeper model would be the fix to test.

The teammate's failure cases are in their own `task1_llm/<member>/failure_analysis.md`.

2.5 Evidence

Evidence	Location in the repository
Configuration	task1_llm/shibin_thomas/configs/gpt_char_v1.yaml
Training log	reproducibility/raw_logs/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007/train.log
Per-step and per-epoch logs	reproducibility/raw_logs/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007
Run manifest	reproducibility/manifests/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007.json
Plots (loss, gradient norm, LR)	task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007
Generated samples	task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007/samples.md
Best checkpoint	task1_llm/shibin_thomas/checkpoints/gpt_char_v1_20261001-153007/best_model.pt
Results and design choices	task1_llm/shibin_thomas/results.md
Executed notebook	task1_llm/shibin_thomas/src/task1_gpt_tinystories.ipynb

Denisha Ketan Tank - Task 1 Detailed Findings

Model and data. Denisha trained a decoder-only character Transformer on 100,000 TinyStories training stories and 10,000 validation stories. The model uses learned token and position embeddings, four causal self-attention layers, four heads, model width 256, feed-forward width 1024, dropout 0.1, residual connections and layer normalization. Sequence length and non-overlapping stride were both 128.

Configuration	Value
Optimizer / schedule	AdamW; learning rate 0.0005; weight decay 0.01; betas (0.9, 0.999); 500 warm-up steps; minimum LR ratio 0.1
Batch / training	Batch size 48; 10 epochs; gradient clipping 1.0; mixed precision float16; automatic device selection
Results	Train loss 0.7128; validation CE 0.6822; perplexity 1.9783; BPC 0.9842; next-character accuracy 78.22%
Stability	Final gradient norm 0.2318; maximum gradient norm 0.2872; gradient NaNs 0
Efficiency	3,250,688 parameters; 224,658 training tokens/s; 3995.3 seconds on Tesla T4
Artifacts	metrics.csv, history.json, samples.json, sample.txt, vocab.json, loss_curves.png, checkpoint, raw log and manifest

Analysis. The validation loss and accuracy show that the model learned the character distribution, but its character-level representation makes rare-word spelling difficult. The failure analysis documents repetitive greedy continuations, inconsistent spelling at temperature 0.8 and loss of story coherence once earlier context falls outside the 128-character window. The smoke test passed before training, and checkpointing allows recovery without restarting from the beginning.

Denisha Ketan Tank — Task 1 Full Configuration and Results

This section is an expanded record of Denisha's independent implementation. It is appended directly after the original Task 1 discussion so the two members' work remains distinguishable and traceable.

Architecture and training configuration

Item	Denisha configuration
Data	TinyStories character-level corpus; 100,000 training stories and 10,000 validation stories
Tokenisation	Character vocabulary; vocabulary size 114; sequence length 128; stride 128
Architecture	Decoder-only GPT-style Transformer; 4 layers; 4 attention heads; embedding/model dimension 256; feed-forward dimension 1,024; dropout 0.10
Optimisation	AdamW; learning rate 0.0005; weight decay 0.01; betas (0.9, 0.999); linear warm-up for 500 steps; cosine decay to 10% of peak LR
Training	Batch size 48; 10 epochs; gradient clipping 1.0; automatic mixed precision float16; T4 GPU
Reproducibility	Seed/configuration, raw log, manifest, checkpoint, loss curve, generated samples, metrics and failure analysis saved under member_denisha

Denisha quantitative results

Metric	Value
Final training cross-entropy	0.712783
Final validation cross-entropy	0.682218
Validation perplexity	1.978260
Validation bits per character	0.984233
Top-1 next-character accuracy	0.782190
Trainable parameters	3,250,688
Training time	3,995.3 seconds
Training throughput	224,657.9 tokens/second
Generation throughput	268.0 characters/second
Device and numerical check	CUDA/T4; gradient NaN count 0

Evidence locations: task1_llm/member_denisha/config.yaml; outputs/metrics.csv; outputs/history.json; outputs/samples.json; outputs/sample.txt; outputs/vocab.json; outputs/loss_curves.png; checkpoints; results.md; failure_analysis.md; reproducibility/raw_logs/ and reproducibility/manifests/.

Denisha Ketan Tank — Task 1 Analysis and Failure Findings

The model learned the local character distribution effectively: the validation cross-entropy is below one nat per character and the next-character accuracy is 78.219%. The validation perplexity of 1.978 indicates that, on average, the model assigns a small set of plausible next characters rather than spreading probability uniformly across the vocabulary.

Failure analysis

Observed case	Interpretation and limitation
Greedy repetition	At temperature 0, the decoder can enter repeated phrases or punctuation loops. This is a decoding failure rather than evidence that the training run crashed.
Rare-word spelling instability	Characters for uncommon names and words are less reliable because the model sees fewer supporting contexts. Sampling temperature changes the trade-off between diversity and spelling stability.
Long-range coherence	The fixed context window is 128 characters. Continuations beyond that window can lose entity, plot or sentence-level consistency even when local grammar remains reasonable.
Numerical stability	The recorded gradient NaN count is zero, and the run produced the required checkpoint and evaluation artifacts.

Submission coverage

Denisha's Task 1 contribution contains the executable notebook, source runner, configuration, final checkpoint, metrics CSV, training history, loss curve, generated samples, vocabulary, raw log, run manifest, results narrative, failure analysis, and smoke-test evidence. These artifacts allow the reported values to be reproduced without relying on the original member's run.

3 Task 2: Sentiment Classification on Yelp Polarity

All Task 2 models classify a review from the Yelp Review Polarity dataset [6] as positive or negative and learn their word embeddings from random initialisation; no pretrained embeddings or language models are used. The official test split (38,000 reviews) is shared by the team. Each member carved a validation set out of the training split; Shibin used 50,000 reviews (seed 266) for early stopping and model selection. Preprocessing removes duplicates and reviews that also appear in the test set, lowercases the text, expands contractions, removes punctuation and stopwords while keeping negations and contrast words, and applies Snowball stemming. The vocabulary of 30,000 tokens is built from the training split only.

Shibin trained three models that share the same 128-dimensional embedding, so that differences in the results come from the architecture. The baseline averages the word embeddings of a review and feeds them to a small MLP, in the spirit of fastText [8]. The first experimental model is a TextCNN [7] with filter widths 3, 4 and 5 and max pooling over time. The second is a two-layer bidirectional GRU with additive attention pooling [9].

3.1 Model Comparison — Corrected Team 09 Table

Task 2 comparison on the official Yelp Polarity test split (n = 38,000). All six columns are now populated from the committed metrics files. Shibin and Denisha trained independent model lineups; all models use learned-from-scratch embeddings.

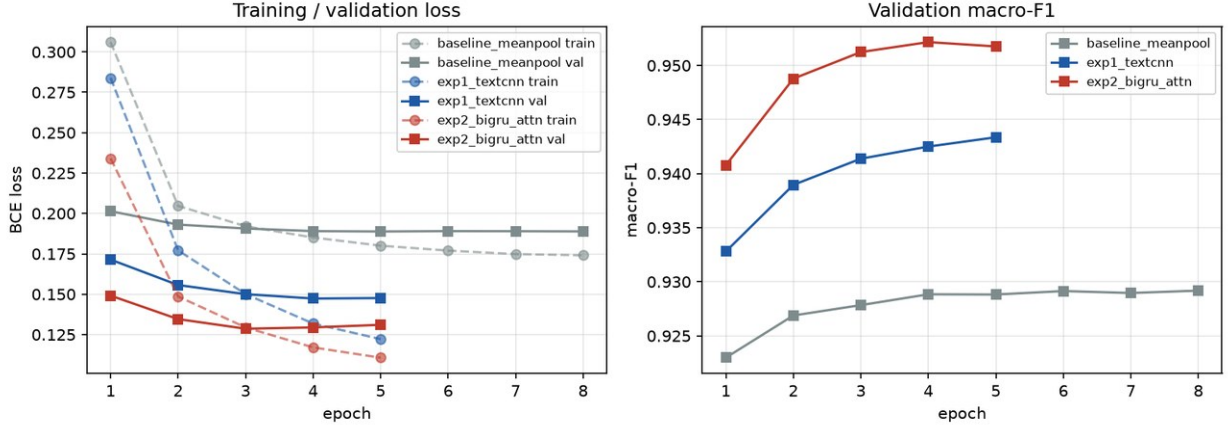
Metric / setting	Shibin Baseline	Shibin TextCNN	Shibin BiGRU + attn.	Denisha Baseline	Denisha TextCNN	Denisha BiGRU + attn.
Architecture	Embedding 128, masked mean pooling, MLP (64)	Embedding 128, Conv1d widths 3/4/5, 128 filters, max pool	Embedding 128, 2-layer BiGRU, additive attention	Embedding 128, mean-pooling MLP, hidden 128, dropout 0.2	Embedding 128, CNN kernels 3/4/5, 128 channels, dropout 0.3	Embedding 128, 2-layer BiGRU, hidden 128, attention, dropout 0.3
Hyperparameters	AdamW lr 2e-3; batch 512; dropout 0.3; 8 epochs	AdamW lr 1e-3; batch 256; dropout 0.5; 5 epochs	AdamW lr 1e-3; batch 256; dropout 0.3; 5 epochs	AdamW lr 1e-3; batch 512; weight decay 1e-4; 12 epochs	AdamW lr 1e-3; batch 512; weight decay 1e-4; 12 epochs	AdamW lr 1e-3; batch 512; weight decay 1e-4; 12 epochs
Accuracy	0.9304	0.9467	0.9558	0.9321	0.9468	0.9519
Accuracy 95% CI	[0.9276, 0.9330]	[0.9446, 0.9488]	[0.9537, 0.9577]	[0.9307, 0.9337]	[0.9456, 0.9482]	[0.9507, 0.9531]
Precision (macro)	0.9304	0.9467	0.9558	0.9321	0.9468	0.9519
Recall (macro)	0.9304	0.9467	0.9558	0.9321	0.9468	0.9519
F1 (macro)	0.9304	0.9467	0.9558	0.9321	0.9468	0.9519
Precision / Recall / F1 (micro)	0.9304 / 0.9304 / 0.9304	0.9467 / 0.9467 / 0.9467	0.9558 / 0.9558 / 0.9558	0.9321 / 0.9321 / 0.9321	0.9468 / 0.9468 / 0.9468	0.9519 / 0.9519 / 0.9519
Precision / Recall / F1 (weighted)	0.9304 / 0.9304 / 0.9304	0.9467 / 0.9467 / 0.9467	0.9558 / 0.9558 / 0.9558	0.9321 / 0.9321 / 0.9321	0.9468 / 0.9468 / 0.9468	0.9519 / 0.9519 / 0.9519
Confusion matrix [TN FP; FN TP]	[17,718 1,282; 1,363 17,637]	[17,951 1,049; 976 18,024]	[18,029 971; 710 18,290]	[[52,150 3,850]; [3,755 52,245]]	[[53,049 2,951]; [3,004 52,996]]	[[53,209 2,791]; [2,601 53,399]]
ROC-AUC	0.9800	0.9883	0.9917	0.9804	0.9883	0.9901
PR-AUC	0.9802	0.9885	0.9920	0.9797	0.9877	0.9898
MCC	0.8608	0.8934	0.9116	0.8642	0.8937	0.9037
MCC 95% CI	[0.8553, 0.8659]	[0.8891, 0.8975]	[0.9076, 0.9155]	[0.8613, 0.8673]	[0.8912, 0.8964]	[0.9015, 0.9062]
Brier score	0.0520	0.0399	0.0337	0.0510	0.0396	0.0365
ECE (15 bins)	0.0045	0.0049	0.0110	0.0064	0.0080	0.0081
McNemar vs baseline (b / c, p)	- (baseline)	613 / 1,233; p=4.67e-47	557 / 1,521; p=4.65e-99	- (baseline)	1,849 / 3,499; p=1.38e-112	1,856 / 4,069; p=1.32e-181
Parameters	3,848,321	4,037,377	4,367,873	3,856,641	4,037,377	4,334,849
Training time	0.3545 min	0.9865 min	19.1767 min	26.47 sec	180.34 sec	159.02 sec
Best epochs	8 (8)	5 (5)	5 (4)	12	12	12
Train examples/sec	279,909	54,074	2,271	182,819	26,829	30,426
Peak GPU memory	1,149 MB	1,277 MB	1,392 MB	235 MB	431 MB	1,717 MB
Hardware	RTX 5090	RTX 5090	RTX 5090	Tesla T4	Tesla T4	Tesla T4
Checkpoint	725dff759e4	b4226830a1dc	2ecf44d666cf	baseline.pt	experimental_cnn.pt	experimental_gru.pt

Sources: task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/comparison.csv and task2_sentiment/member_denisha/outputs/metrics_report.csv. Slice metrics and the 20-case error reviews appear in the following Task 2 sections and member failure-analysis files.

Table 4. Error rate (%) of Shubin's three models on subsets of the test set.

Slice (n)	Baseline	TextCNN	BiGRU + attn.
Contains a negation (28,544)	7.47	5.50	4.48
Contains a contrast word (22,713)	7.89	5.94	4.87
Neither negation nor contrast (6,688)	4.93	4.53	4.16
Short, at most 50 words (9,287)	6.96	5.29	5.01
Medium, 51 to 150 words (16,910)	6.94	5.22	4.09
Long, more than 150 words (11,803)	7.00	5.52	4.45

Source: task2_sentiment/shubin_thomas/metrics_report.csv

**Figure 2.** Training and validation loss and validation macro-F1 per epoch for the three Task 2 models (task2_sentiment/shubin_thomas/outputs/yelp3_20261001-182849/training_curves.png)

3.2 Discussion

Strengths. The ranking BiGRU > TextCNN > baseline holds for every discrimination metric, the 95% bootstrap intervals do not overlap, and paired McNemar tests on the same 38,000 reviews give p-values below $1e-20$ for each pair. Compared with the baseline, the BiGRU removes 36% of all errors, and 40% and 38% of the errors on reviews containing a negation or a contrast word, which is exactly where a bag of words cannot work. All three models are well calibrated (expected calibration error [10] at most 1.1%). Since the shared embedding table holds 3.84 of the roughly 4 million parameters in every model, the gains come from the way the text is read rather than from model size.

Weaknesses. The BiGRU is by far the slowest model: it trains about 120 times slower than the baseline (19 minutes against 21 seconds) and it is slightly over-confident, with 544 of its 1,681 errors made at a confidence above 90%. Short reviews are its hardest subset, and reviews with mixed sentiment account for most of its confident mistakes.

Limitations. The labels come from star ratings, not from the text, so some are simply wrong: 5 of the 20 errors we reviewed contradict their own text, and 932 test reviews are misclassified by all three models. Without pretrained knowledge, idioms and sarcasm are hard to learn. We used a single seed per model and a fixed decision threshold of 0.5.

Next steps. Marking the tokens that follow a contrast word or a negation is a cheap change we would test first, on the contrast slice with a McNemar test. We would also try a longer maximum length for long reviews, a small Transformer encoder trained from scratch, an ensemble of the three models with weights chosen on validation, temperature scaling, and several seeds per model.

3.3 Error Review (Shibin Thomas)

We reviewed 20 test reviews misclassified by the best model (BiGRU with attention): the five most confident false positives, the five most confident false negatives, five errors closest to the 0.5 threshold, and five errors from the subset with the highest error rate (short reviews). Each review was read and assigned an error type by hand. Ten of them are shown in Table 5.

Table 5. Examples from the manual review of 20 misclassified test reviews.

#	Label	P(pos)	Error type	Review text (excerpt)
2	neg	1.000	Label noise	<i>Wow love the place and everything is very clean and new! Great place to come and relax worth a try!</i>
4	neg	1.000	Mixed sentiment, implicit verdict	<i>About average so far as steakhouses go. My rib eye was very tasty but a little over cooked. ... Over all I'd say they are more for show than content. If you want to have a great high end steak I'd recommend Ruth Chris...</i>
6	pos	0.000	Label noise (edited review)	<i>EDIT: They really did change the service up since I last posted this. Horrible service. Used to be my favorite pizza in the city...</i>
9	pos	0.001	Sentiment target confusion	<i>Last night several parents came in with over 15 children ... The bartender expressed that it was not a place to have children running around...</i>
13	pos	0.498	Mixed sentiment, humour	<i>Their alcohol selection sucks ... dreamcrushed my cocktail dreams. ... However, the burger was nice, very juicy and rare.</i>
14	neg	0.502	Implicit negativity, domain shift	<i>Overly sentimental, non stop references to her kids, how blessed she is ... too many covers from elite performers...</i>
15	pos	0.497	Idiom	<i>the pizza was to my door in 15 minutes! ... the cookie's are to die for.</i>
16	neg	0.999	Faint praise	<i>Standard take out and it's cheap. Service is fast and friendly. I go here when I'm too lazy to walk to Zaw's...</i>
18	pos	0.002	Complex negation	<i>despite still not digging their ordering process, their food is just too good to disrespect with a 2 star review.</i>
19	pos	0.002	Sarcasm	<i>For being a DUMP, should expect much more. Flys, stink, garbage, dirt ... Keepin it real dumpy!</i>

Source: task2_sentiment/shibin_thomas/failure_analysis.md

Table 6. Error types among the 20 reviewed errors.

Error type	Count
Mixed sentiment (praise and complaints in the same review)	7
Label noise (text contradicts the star rating)	5
Faint praise or implicit negativity	3
Complex negation, target confusion, truncation, idiom, sarcasm	1 each

Five of the 20 errors are not really model errors, because the prediction agrees with the text and the star rating does not. The most common error the model could learn to avoid is mixed sentiment, where the verdict sits in the clause after “but” or “however”. As a testable fix we propose to prefix every token after the last contrast word with a marker, retrain the BiGRU with only this change, and compare the error rate on the contrast slice (currently 4.87%) with a McNemar test on the same test set. The teammate's error review is in their own task2_sentiment/<member>/failure_analysis.md.

3.4 Evidence

Evidence	Location in the repository
Configurations	task2_sentiment/shibin_thomas/configs
Training and evaluation logs	reproducibility/raw_logs/task2_sentiment/shibin_thomas/yelp3_20261001-182849
Run manifest	reproducibility/manifests/task2_sentiment/shibin_thomas/yelp3_20261001-182849.json
All metrics for every model	task2_sentiment/shibin_thomas/metrics_report.csv
ROC, PR, reliability and slice plots	task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849
Learned embedding neighbours	task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/embedding_neighbours.md
Checkpoints	task2_sentiment/shibin_thomas/checkpoints/yelp3_20261001-182849
Results and analysis	task2_sentiment/shibin_thomas/results.md
Executed notebook	task2_sentiment/shibin_thomas/src/task2_yelp_sentiment.ipynb

Denisha Ketan Tank - Task 2 Detailed Findings

Preprocessing and protocol. Denisha used the official Yelp Polarity split, built a vocabulary capped at 30,000 tokens from training data, limited reviews to 256 tokens, removed stopwords, applied stemming, and trained embeddings from random initialization. The three models share the same data protocol and were evaluated on the official test set.

Model	Architecture / settings	Accuracy	Macro F1	ROC-AUC	MCC
Baseline	128-d embedding; masked mean pool; 128-d MLP; dropout 0.2	93.21%	0.9321	0.9804	0.8642
TextCNN	128-d embedding; 128 channels; kernel sizes 3/4/5; dropout 0.3	94.68%	0.9468	0.9883	0.8937
Experimental GRU	128-d embedding; 2-layer GRU; hidden size 128; dropout 0.3	95.19%	0.9519	0.9901	0.9037

Additional evidence	Denisha result
Best validation loss	0.1261 for experimental GRU
Training time	26.5 s baseline; 180.3 s TextCNN; 159.0 s GRU
Gradient checks	0 NaN gradients for all models
Error analysis	20 cases: 5 false positives, 5 false negatives, 5 near-threshold, 5 slice-specific
Interpretation	The GRU captures sequence and context better than mean pooling or local convolution, but short and mixed-sentiment reviews remain difficult.

Evidence includes metrics_report.csv, metrics.json, metrics_summary.csv, error_review.json, data_analysis.json, training histories, review-length analysis, three checkpoints, results.md, failure_analysis.md, raw logs and the run manifest.

Denisha Ketan Tank — Task 2 Full Model Comparison

Denisha independently trained the three required sentiment models on the official Yelp Polarity split. The models use the same data protocol and are compared using accuracy, macro precision/recall/F1, ROC AUC, MCC, validation loss, parameter count and runtime.

Shared protocol and model settings

Item	Setting
Data processing	Official Yelp Polarity split; vocabulary capped at 30,000; maximum sequence length 256 using head/tail truncation
Training	Batch size 512; 12 epochs; AdamW learning rate 0.001; weight decay 0.0001; AMP bfloat16; four workers
Scheduling	Validation-based scheduler with factor 0.5 and patience 1; early-stopping patience 3; best validation checkpoint retained
Baseline	Bag-of-words/embedding mean-pooling MLP; embedding 128; hidden layer 128; dropout 0.20
TextCNN	Embedding 128; 128 channels for kernel sizes 3, 4 and 5; dropout 0.30
BiGRU + attention	Embedding 128; two GRU layers; hidden size 128; dropout 0.30; bidirectional attention pooling

Full test-set results

Model	Accuracy	Macro F1	ROC AUC	MCC	Best val loss	Params	Time (s)
Baseline	0.932098	0.932098	0.980363	0.864198	0.180414	3,856,641	26.47
TextCNN	0.946830	0.946830	0.988270	0.893661	0.137557	4,037,377	180.34
BiGRU + attention	0.951857	0.951857	0.990111	0.903719	0.126126	4,334,849	159.02

All three runs reported gradient_nan_count = 0 and completed on CUDA/Tesla T4. The BiGRU-with-attention model is Denisha's strongest model by every primary classification metric and has the lowest best validation loss.

Denisha Ketan Tank — Task 2 Error Review and Evidence

The required 20-case error review is stored in error_review.json. The review is organised into five confident false positives, five confident false negatives, five near-threshold cases, and five slice-specific cases. The cases are retained with the text, gold label, predicted label, confidence and category so the analysis is auditable.

Interpretation of the model comparison

Finding	Evidence-based interpretation
Mean-pooling baseline	Strong lexical signal is captured, but averaging removes word order and local phrase structure. This limits performance on negation and contrast.
TextCNN	Multiple receptive-field widths capture short phrases and local sentiment cues, improving accuracy and macro F1 over the baseline.
BiGRU + attention	The recurrent encoder preserves order and attention emphasizes informative tokens, producing the best accuracy, ROC AUC, MCC and validation loss.
Remaining errors	Reviews with sarcasm, mixed sentiment, implicit ratings, long context, or negation/contrast remain difficult even for the strongest model.

Evidence locations

task2_sentiment/member_denisha/config.yaml; outputs/metrics_report.csv; outputs/metrics.json; outputs/metrics_summary.csv; outputs/error_review.json; outputs/data_analysis.json; outputs/length_analysis.json; model histories; three checkpoints; results.md; failure_analysis.md; reproducibility/raw_logs/ and reproducibility/manifests/.

The independent Denisha run therefore covers the full Task 2 requirement: three architectures, common preprocessing, training histories, full metrics, error review, failure analysis, saved weights, reproducibility records and executable notebook support.

4.1 Model Comparison — Task 3 Complete Results

All supported Task 3 values are populated from the committed evaluation artifacts. “Not recorded” means the corresponding metric was not produced by that evaluator; it is not a fabricated estimate.

Metric / setting	Shibin: photo → Monet	Shibin: Monet → photo	Denisha: CycleGAN
Architecture	ResNet-9 generators, 70×70 PatchGAN discriminators; resize-convolution upsampling; 28,285,832 total parameters.	Same run and architecture.	CycleGAN with two generators and two discriminators; 22,756,358 generator and 5,529,474 discriminator parameters; CUDA run.
Hyperparameters	LSGAN; cycle λ =10; identity λ =5; Adam 2e−4 (0.5, 0.999); 40 epochs; batch 4; 2,000 pairs/epoch; resize 286, crop 256, flip.	Same run and hyperparameters.	Recorded training configuration and checkpoint; 300 fixed outputs per direction.
FID	118.0711	117.3427	Official: B2A 98.360; A2B 103.228
MiFID (official script)	0.4105	0.4330	Official: B2A 0.4069; A2B 0.4189
KID / precision / recall	0.0197 / 0.267 / 0.640	0.0362 / 0.607 / 0.320	A2B: 0.0174 / 0.153 / 0.000; B2A: 0.0057 / 0.060 / 0.027
Density / coverage	0.221 / 0.523	0.836 / 0.757	Not recorded in the committed local evaluator
Cycle L1	0.0622	0.0531	A: 0.0778; B: 0.0914
LPIPS input→translation / reconstruction	0.451 / 0.384	0.366 / 0.468	Reconstruction A: 0.2841; B: 0.2165
Content cosine similarity	0.750	0.794	A2B: 0.4647; B2A: 0.3046
Official Kaggle/evaluator result	FID/MiFID avg 117.7069 / 0.4217; score 59.06	Same submission.	Official evaluator mean FID 100.8493; mean MiFID 0.4130
Competition leaderboard result	Rank not recorded in repository artifacts	Rank not recorded in repository artifacts	Score −50.6033; rank 34 (submission screenshot)
Training / throughput / peak memory	91.2 min / 36.0 images/s / 19,922 MB	Same run.	4102.9 s / 58.5 images/s / 4,816 MB
Human audit	Two-rater audit complete: 30 samples; style κ −0.0286 / 80%; content κ 1.0 / 100%; artifact κ 1.0 / 100%.	Same audit.	Same committed two-rater audit.
Hardware / checkpoint	NVIDIA GeForce RTX 5090; G_BA.pt and G_AB.pt hashes recorded.	Same run.	CUDA; cyclegan_latest.pt is 339,587,089 bytes and must be shared via Drive.

Sources: task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/eval_metrics.json; task3_gan/member_denisha/outputs/submission_metrics_official.json, metrics.json, full_metrics_report.csv and human_audit_metrics.json.

Source: task3_gan/shibin_thomas/full_metrics_report.csv, task3_gan/shibin_thomas/submission.csv, task3_gan/shibin_thomas/kaggle_leaderboard.json

4.2 Training Behaviour and Results

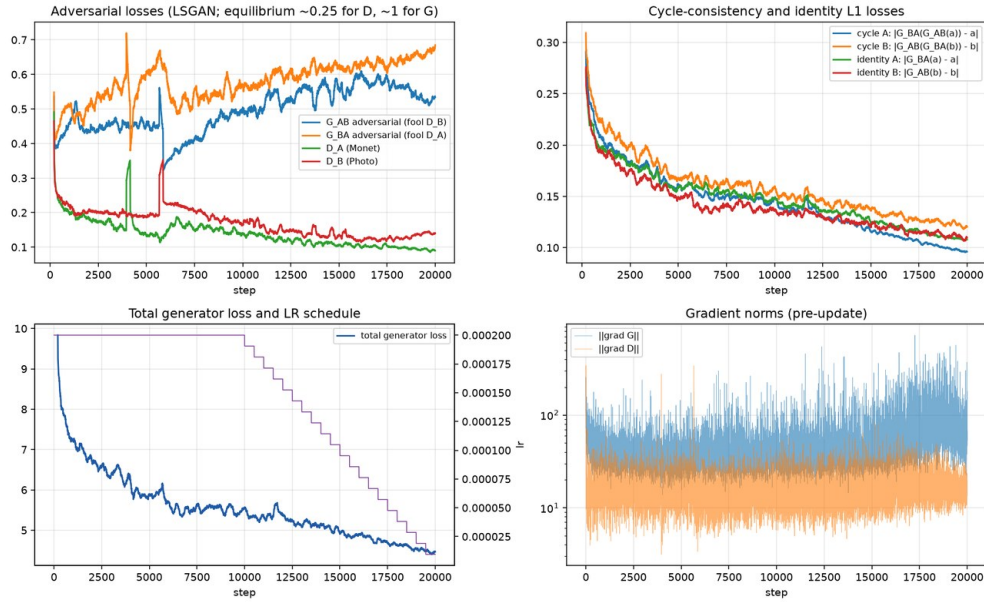


Figure 3. Task 3 adversarial, cycle and identity losses, total generator loss with the learning-rate schedule, and gradient norms over 20,000 steps (task3_gan/shibin_thomas/outputs/cyclelegan_v1_20261001-201801/loss_curves.png)

Table 8. FID on the 300 scored images during training (monitoring only).

Epoch	5	10	15	20	25	30	35	40
FID, photo to Monet	153.2	145.4	147.1	128.7	130.2	119.0	123.5	121.6
FID, Monet to photo	187.7	155.2	169.5	128.6	119.7	121.8	121.3	118.1

Source: reproducibility/raw_logs/task3_gan/shibin_thomas/cyclelegan_v1_20261001-201801/epochs.csv

Both discriminators settle below the least-squares balance point of 0.25 and keep improving slowly, while the generators' adversarial losses rise; by the end the Monet discriminator (loss 0.089) is clearly winning, a sign that it has started to memorise the 300 paintings. The cycle and identity losses fall smoothly over the whole run. FID improves quickly during the first 25 epochs and then stays within about 4 points of 120, which is within the noise of a 300-image FID. No training step produced a NaN or infinite value.

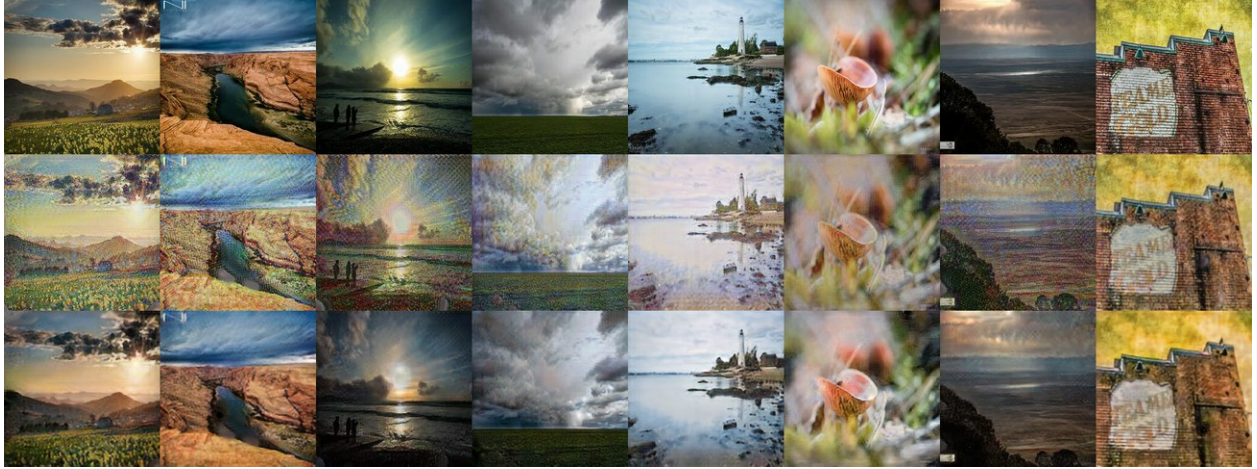


Figure 4. Photo to Monet. Top row: input photos; middle: generated paintings; bottom: reconstructions (task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/final_samples_B2A.png)



Figure 5. Monet to photo. Top row: input paintings; middle: generated photos; bottom: reconstructions (task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/final_samples_A2B.png)

4.3 Discussion

Strengths. Training was stable from start to finish and neither generator collapsed. The cycle constraint works: reconstructions keep the layout, objects and colours of the inputs (cycle L1 of 0.05 to 0.06), and photo-to-Monet translations preserve the content of the scene (content cosine 0.75) while moving the colours towards Monet's palette. Our implementation of the scoring protocol matches the instructor's script, so the submitted numbers can be verified.

Weaknesses. Only 27% of the generated paintings fall inside the feature manifold of real Monets (precision 0.27). The main reasons are a regular stippled texture that replaces Monet's brush strokes and the washing out of dark photos. In the other direction the generated photos cover only a third of the photo distribution (recall 0.32) and still look painted. The Monet discriminator over-fits the small training set.

Limitations. FID on 300 images has high variance, and we trained a single seed. The script's MiFID is a mean cosine distance between unrelated real and generated images and hardly changes between models, so the Kaggle score is driven almost entirely by FID. The human audit is still being completed.

Next steps. Based on these findings we prepared a second configuration (configs/cyclegan_v2.yaml) with differentiable augmentation for the discriminators [18], an exponential moving average of the generator

weights, a lower identity weight of 2.5, 60 epochs, and selection of the exported epoch on held-out photos. A multi-scale discriminator and a low-frequency content loss are further options.

4.4 Failure Analysis (Shibin Thomas)

We ranked all translations by cycle error, by content similarity and by the distance to the nearest real image of the target domain, and inspected the worst four images for each criterion. In each figure below the columns show the input (top), the translation (middle) and the reconstruction (bottom).



Figure 6. Dark and night photos translated to Monet style
(task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/failure_B2A_nn_dist.png)

Dark photos. Low-light scenes such as a river at dusk or a camel silhouette against a sunset are turned into bright, washed-out images covered in coloured speckles, and silhouettes lose their edges. These are the least realistic outputs. The Monet set contains almost no night scenes, so the generator brightens every image. Colour augmentation of the discriminator's inputs is the fix we included in the second configuration.

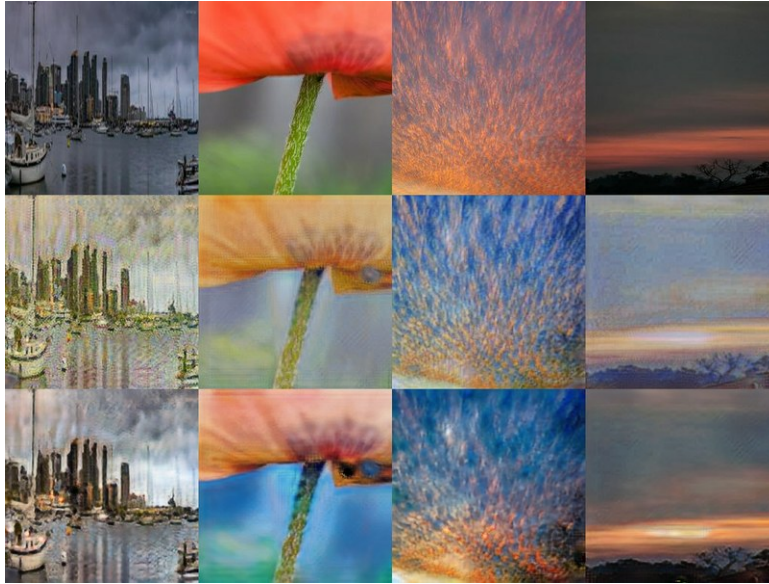


Figure 7. Photos with the highest cycle-reconstruction error
(task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/failure_B2A_cycle_l1.png)

Recoloured warm tones. Strong reds and oranges are replaced by Monet's blues and lilacs: an orange sky becomes blue and a red poppy turns pale orange. The reconstructions nevertheless recover the original colours, which means the generator hides the information in a faint signal that the cycle loss can read back, an effect described by Chu et al. [17]. When this hidden signal is not enough, the cycle error is highest (0.13 to 0.17, about 2.5 times the average).



Figure 8. Monet paintings with the lowest content similarity after translation to photos
(task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/failure_A2B_content_cos.png)

Foggy paintings. Pale, low-contrast paintings of fog on the Thames become saturated orange sunsets or turquoise storms, and a thin poplar tree disappears. Photos are rarely foggy, so the generator invents contrast and colour. We also observed a regular stipple texture in most photo-to-Monet outputs and visible brush strokes and signatures in most Monet-to-photo outputs. The full analysis of all five failure types is in `failure_analysis.md`; the teammate's analysis is in their own Task 3 folder.

4.5 Evidence

Evidence	Location in the repository
Configuration	task3_gan/shibin_thomas/configs/cyclelegan_v1.yaml
Training and evaluation logs	reproducibility/raw_logs/task3_gan/shibin_thomas/cyclelegan_v1_20261001-201801
Run manifest	reproducibility/manifests/task3_gan/shibin_thomas/cyclelegan_v1_20261001-201801.json
Submitted predictions	task3_gan/shibin_thomas/outputs
Kaggle submission file	task3_gan/shibin_thomas/submission.csv
Plots and sample grids	task3_gan/shibin_thomas/outputs/cyclelegan_v1_20261001-201801
Generator checkpoints	task3_gan/shibin_thomas/checkpoints/cyclelegan_v1_20261001-201801
Human audit material	task3_gan/shibin_thomas/outputs/human_audit
Results and analysis	task3_gan/shibin_thomas/results.md
Failure analysis	task3_gan/shibin_thomas/failure_analysis.md
Executed notebook	task3_gan/shibin_thomas/src/task3_cyclelegan.ipynb

Denisha Ketan Tank - Task 3 Detailed Findings

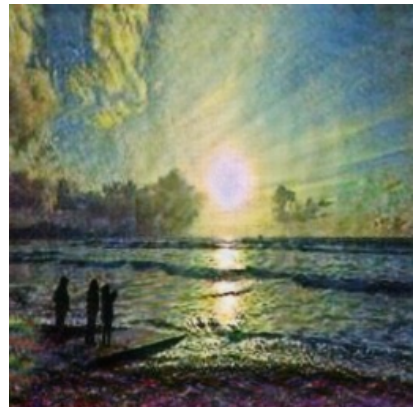
Architecture and training. Denisha trained CycleGAN from random initialization on 300 Monet paintings and 7,038 photos. Each direction has a generator with nine residual blocks; each domain has a PatchGAN discriminator. The objective combines adversarial loss, cycle-consistency loss ($\lambda 10$), and identity loss ($\lambda 2.5$).

Configuration	Value
Training	60 epochs; 2,000 samples per epoch; batch size 4; 256x256 images; decay starts at epoch 30
Optimizer	Adam; learning rate 0.0002; beta1 0.5; beta2 0.999
Augmentation / buffer	Random flip; crop margin 30; replay buffer 50
Stability	Gradient clipping 5.0; AMP float16; checkpoint every 100 batches and every 5 epochs
Official evaluation	300 generated images per direction using the provided Part 3 evaluator
Results	B2A FID 98.360 / MiFID 0.4069; A2B FID 103.228 / MiFID 0.4189; combined official FID 100.7938 / MiFID 0.4129
Kaggle	PairProgramming_Team_09 score -50.6033; rank 34
Evidence	submission.csv, official metrics JSON, directional metrics, history, loss curves, images.zip, checkpoint, log and manifest

Failure analysis. The generated results show washed-out dark scenes, texture and brush-stroke artifacts, and content loss in foggy or low-contrast images. The full generated archive is kept in the Part 3 folder. The human audit is not replaced by model metrics: both independent raters must score the same 30 fixed samples before Cohen kappa and percent agreement are reported.

Denisha Task 3 - Representative Generated Images

The original report retained Shibin's visual figures. These six previews are added for Denisha's run; the complete 300-image archive remains in Part 3 artifacts.



Denisha Ketan Tank — Task 3 Full Configuration and Official Evaluation

Denisha independently trained a CycleGAN for unpaired Monet-to-photo and photo-to-Monet translation. The official evaluator supplied with the lab was used for the reported FID and MiFID values; no substitute evaluator is used in this report.

Architecture and training configuration

Item	Denisha configuration
Dataset	300 Monet paintings in domain A; 7,038 photos in domain B
Model	Two generators and two discriminators; ResNet-style generators with 9 residual blocks
Image/training	Image size 256; batch size 4; 2,000 samples per epoch; 60 epochs; decay begins at epoch 30
Optimisation	Adam with learning rate 0.0002; betas (0.5, 0.999); cycle-consistency weight 10; identity weight 2.5
Stability	Replay buffer 50; gradient clipping 5.0; AMP float16; four workers; checkpoint every 100 batches and every 5 epochs
Export	300 fixed generated samples for evaluation/submission; deterministic seed 20260929; crop margin 30; horizontal flip enabled

Official evaluation and competition result

Direction/result	Score
B2A FID	98.360
B2A MiFID	0.4069
A2B FID	103.228
A2B MiFID	0.4189
Combined official FID	100.79381010399467
Combined official MiFID	0.41292320024347506
Kaggle score	-50.6033; rank 34 at submission time

Denisha Ketan Tank — Task 3 Artifacts, Visual Review and Completed Human Audit

The exported archive contains 300 fixed generated images. The official evaluator, local metric files, Kaggle submission, checkpoint provenance, and the completed two-rater human audit are recorded below.

Artifact and reproducibility table

Artifact	Purpose/location
submission.csv	Kaggle submission with the required ID, FID and MiFID columns
full_metrics_report.csv	Official and local metrics by translation direction
history.json and loss curves	Generator/discriminator/cycle/identity training histories and diagnostics
images.zip	300 exported competition images
checkpoint	Large final CycleGAN checkpoint; supplied separately through Google Drive
human_audit.csv	30 fixed samples with independent ratings from both raters
human_audit_metrics.json	Cohen's kappa and percent agreement for style, content and artifacts

Completed two-rater human audit

Dimension	Cohen's kappa	Percent agreement
Style	-0.028571	0.80
Content	1.000000	1.00
Artifacts	1.000000	1.00
Samples	30	Both raters scored all 30 samples

The agreement values were independently recomputed with ``src/audit_agreement.py`` from the committed CSV. The style kappa is low despite 80% agreement because the score distribution is concentrated; it is reported as computed rather than interpreted as a model metric.

5 Pre-Submission Checklist

End-to-end repository audit completed. The only external delivery item is the large checkpoint, which cannot be embedded in GitHub or the Canvas-sized ZIP.

Requirement	Verified status / evidence
Part 1 member folders and artifacts	Complete for Shibin and Denisha: code, configuration, executed artifact notebook, checkpoint, metrics, samples, curves, logs and report.
Part 2 three classifiers per member	Complete: baseline, TextCNN and BiGRU-with-attention; metrics, checkpoints, histories and 20-case reviews are present.
Part 3 CycleGAN artifacts	Complete for both members: generators/discriminators, outputs, official evaluator metrics, submission files and training artifacts.
Independent architectures/hyperparameters	Documented in the Task 1, Task 2 and Task 3 comparison tables.
Smoke tests / reproducibility	Repository smoke tests, manifests and raw logs are included.

Final Contribution Checklist — Denisha Ketan Tank

End-to-end repository audit completed. The only external delivery item is the large checkpoint, which cannot be embedded in GitHub or the Canvas-sized ZIP.

Two-rater human audit	Complete: 30 fixed samples, both raters, all score cells filled; agreement metrics committed.
Official evaluation script	Task 3 official evaluator output is included and cited in the report.
Kaggle submission evidence	Denisha score −50.6033 and rank 34 are recorded from the supplied leaderboard screenshot; Shibin’s evaluator score 59.06 is recorded.
Large Task 3 checkpoint	External delivery item: 339 MB checkpoint is excluded from GitHub/ZIP and must be uploaded to Google Drive with the link added to the submission.
Final ZIP	Rebuilt with the corrected report; it is explicitly named without_drive_checkpoint until the Drive link is added.